



UNIVERSITÀ DI PAVIA
Dipartimento di
Ingegneria Industriale
e dell'Informazione

Course of Robot Control

Prof. Antonella Ferrara



Sixth Part

Outline of the Course

Basic Terminology and Concepts

Reference Frames and Rotation Matrices

Robot Direct Kinematics: the Geometric Approach

Robot Direct Kinematics: the Denavit-Hartenberg Convention Based Approach

Robot Inverse Kinematics

Robot Differential Kinematics

Robot Dynamic Model

Robot Control



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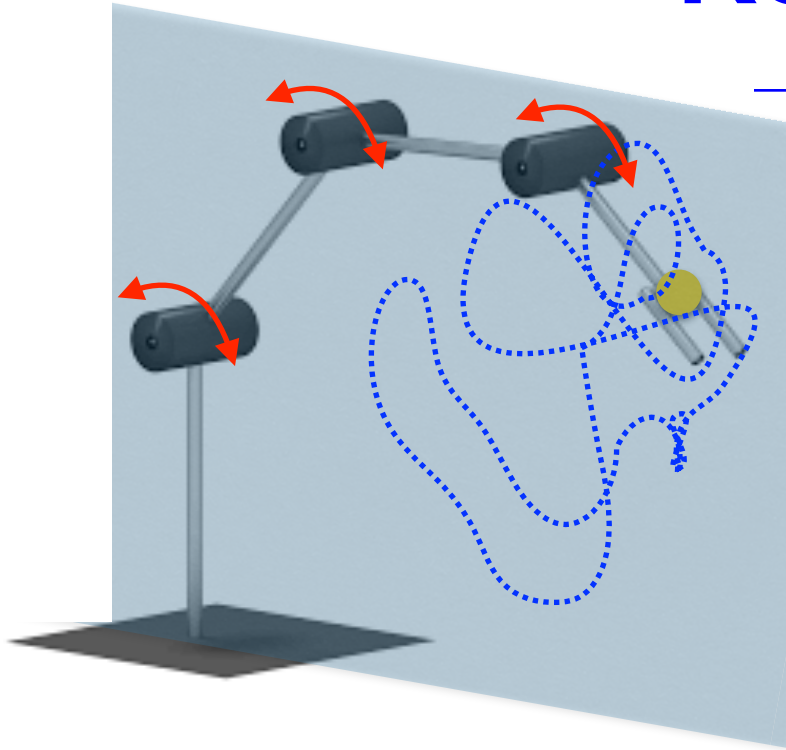
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Robot dynamics and dynamic model formulation

Robot Dynamic Model



Dynamic model of the robot:

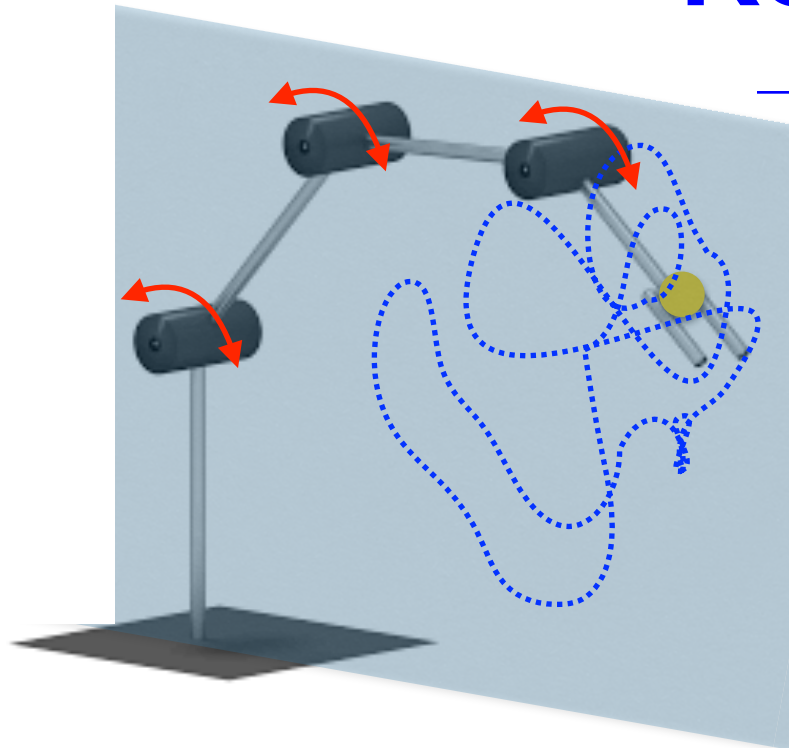
collection of the equations of motion of the manipulator in terms of joint variables and their time derivatives up to the second order

**Joints Torques
or Forces**



**Motion of the
robot ?**

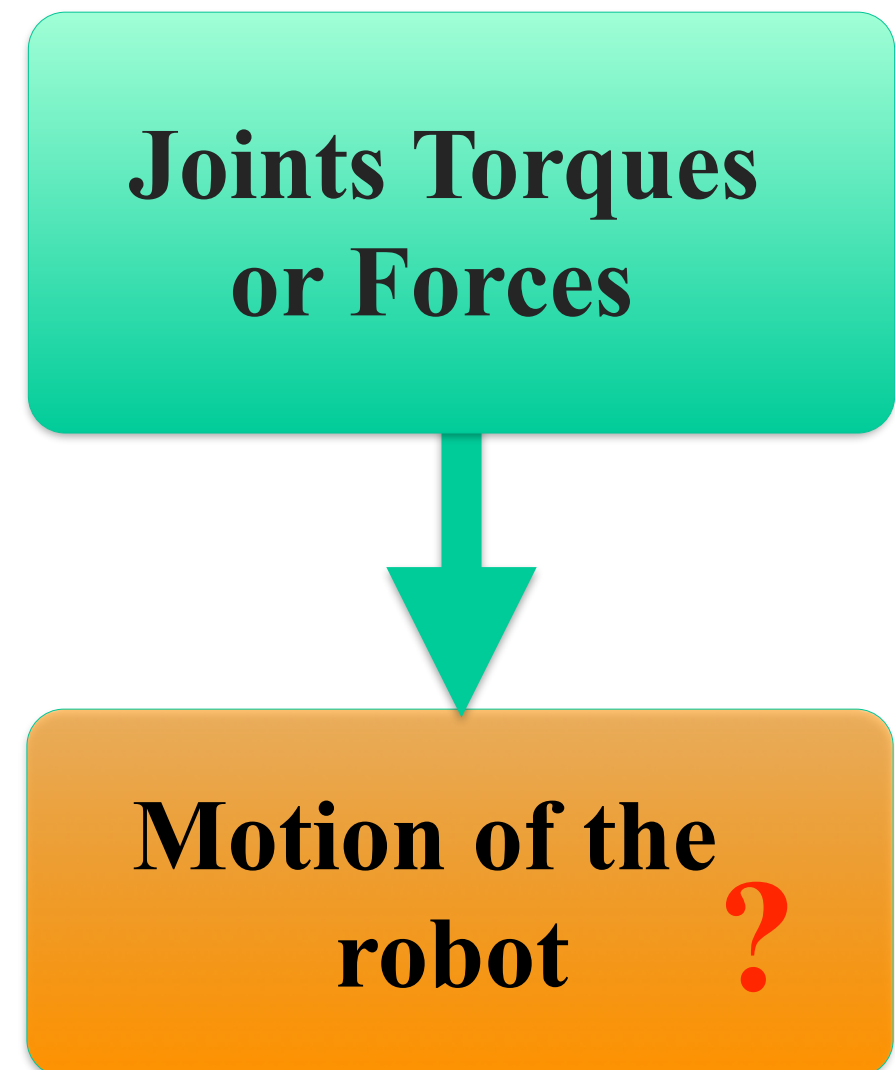
Robot Dynamic Model



The formulation of the dynamic model of a robot manipulator is fundamental **to design the control algorithms**, but it is also important for:

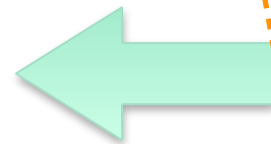
- **motion simulation**
- **motion analysis**
- **performance assessment**

The analysis of the dynamic model can help even in the **mechanical design** of the robot, giving the possibility of a priori determining the forces and torques required for the execution of the robot tasks.



Robot Dynamic Model

Two methods



**Joints Torques
or Forces**



**Motion of the
robot ?**

Robot Dynamic Model

Two methods

Lagrange formulation: conceptually simple and systematic; model in closed form

Newton–Euler formulation: computationally more efficient model in a recursive form

**Joints Torques
or Forces**

**Motion of the
robot ?**

Robot Dynamic Model

Two methods

Lagrange formulation: conceptually simple and systematic; model in closed form

Newton–Euler formulation: computationally more efficient model in a recursive form

**Joints Torques
or Forces**

**Motion of the
robot ?**

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

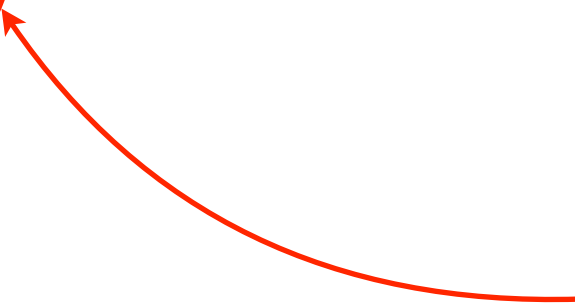
$$q_i, \quad i = 1, \dots, n$$

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$

total kinetic energy



Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$


total potential energy

total kinetic energy

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$

Third Step:

Determine the **Lagrange Equations**

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Second Step:

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**generalized forces
associated
with the generalized
coordinates**



Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

the joint actuator torques/forces, the joint friction torques/forces, as well as the joint torques/forces induced by end-effector forces at the contact with the environment

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$

**nonconservative
generalized
forces**

Third Step:

Determine the **Lagrange Equations**

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

**generalized forces
associated
with the generalized
coordinates**

Robot Dynamic Model

The Lagrange Equations establish the relations existing between:

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

generalized forces applied to the manipulators

**Joints Torques
or Forces**

**Motion of the
robot ?**

joint positions
joint velocities and
joint accelerations

Robot Dynamic Model

The Lagrange Equations establish the relations existing between:

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$



Robot Dynamic Model

generalized forces applied to the manipulators

**Joints Torques
or Forces**



**Motion of the
robot ?**

joint positions
joint velocities and
joint accelerations

Robot Dynamic Model

By using the Lagrange formulation, the robot manipulator dynamic model can be derived in a systematic way independently of the reference coordinate frame which is adopted

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$



Robot Dynamic Model

generalized forces applied to the manipulators

**Joints Torques
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**Motion of the
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Dynamic Model via the Lagrange formulation

Robot Dynamic Model: Lagrange Formulation

First Step:

Select the joint variables (**generalized coordinates**)

$$q_i, \quad i = 1, \dots, n$$

Second Step:

Determine the **Lagrangian** of the mechanical system

$$\mathcal{L} = \mathcal{T} - \mathcal{U}$$

Third Step:

Determine the **Lagrange Equations**

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model: Lagrange Formulation

First Step:

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Third Step:

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$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i$$

$$i = 1, \dots, n$$



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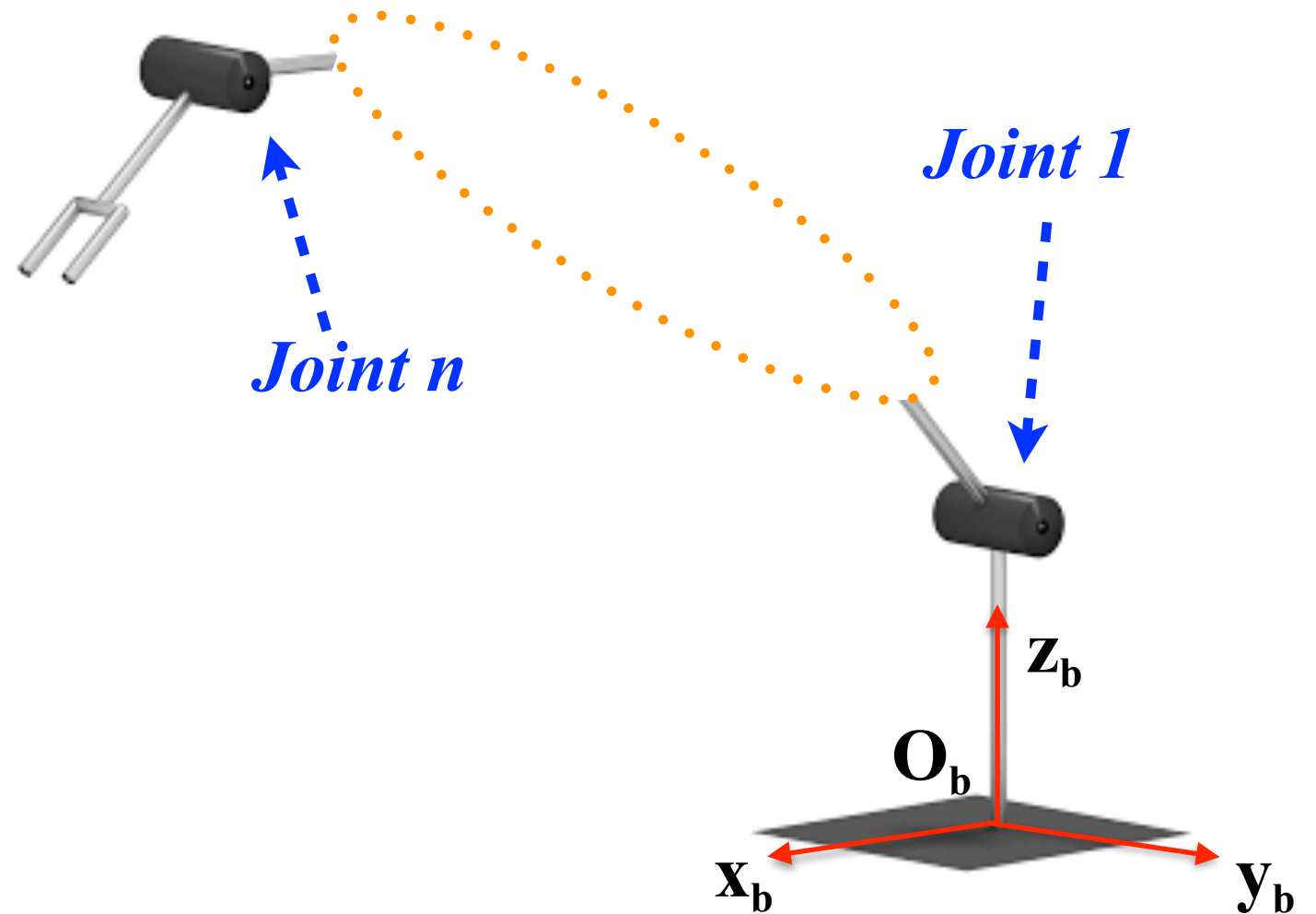


Computation of the Kinetic Energy

Computation of the Kinetic Energy

The total kinetic energy is given by the sum of the contributions relative to the motion of each link and the contributions relative to the motion of each joint actuator (motor)

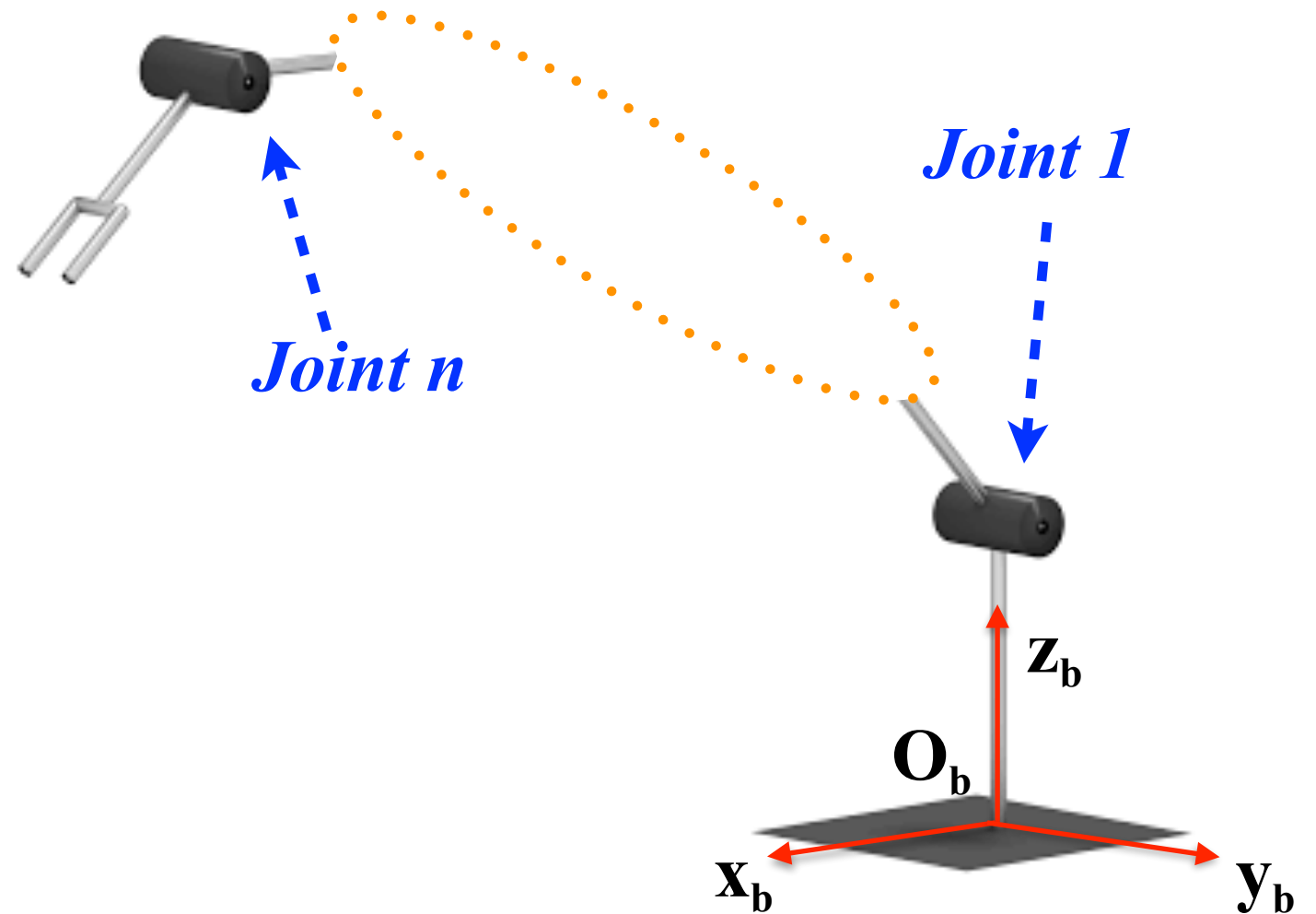
$$T = \sum_{i=1}^n (T_{\ell_i} + T_{m_i})$$



Computation of the Kinetic Energy

The total kinetic energy is given by the sum of the contributions relative to the motion of each link and the contributions relative to the motion of each joint actuator (motor)

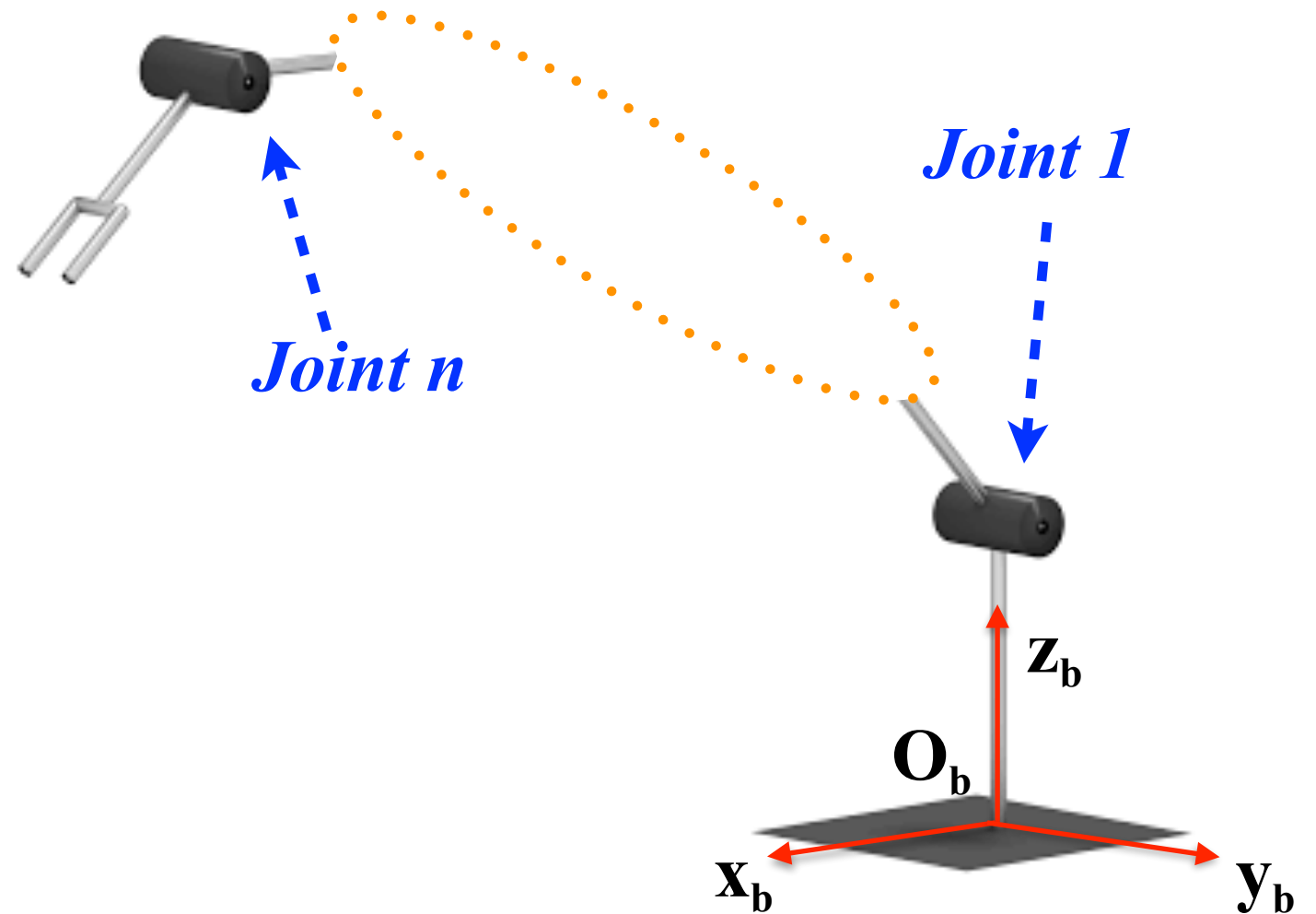
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Computation of the Kinetic Energy

The total kinetic energy is given by the sum of the contributions relative to the motion of each link and the contributions relative to the motion of each joint actuator (motor)

$$T = \sum_{i=1}^n (T_{\ell_i} + T_{m_i})$$



Computation of the Kinetic Energy

Link i

center of mass

elementary particle

density of the elementary particle of volume dV

position vector of the elementary particle

position vector of the link centre of mass w.r.t. base ref frame:

link mass

volume of Link i

velocity vector of link elementary particle

$$T = \sum_{i=1}^n (T_{\ell_i} + T_{m_i})$$

$$T_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \dot{\underline{p}}_i^{*T} \dot{\underline{p}}_i^* \rho dV$$

$$\underline{p}_{\ell_i} = \frac{1}{m_{\ell_i}} \int_{V_{\ell_i}} \underline{p}_i^* \rho dV$$

Computation of the Kinetic Energy

Link i

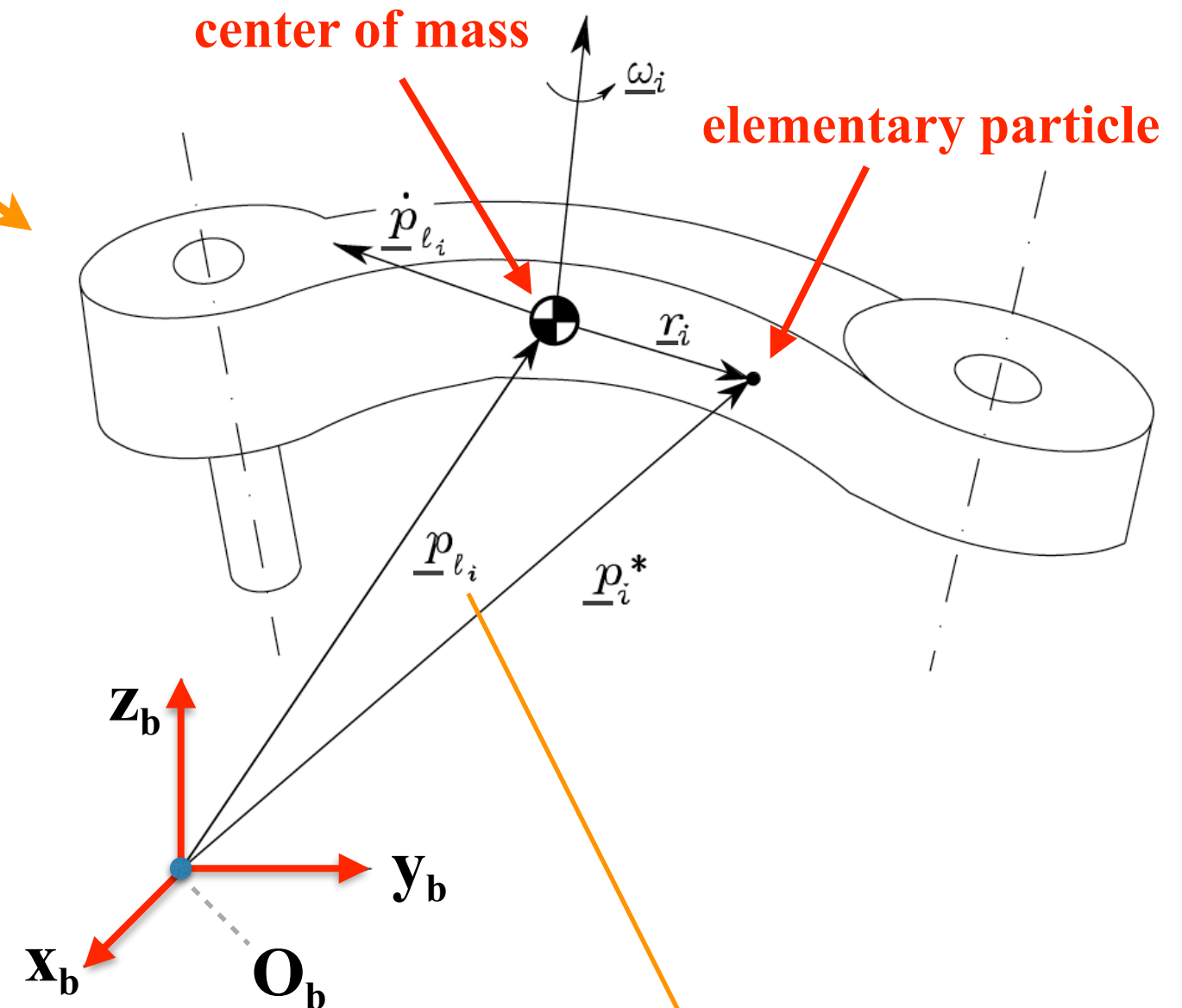
$$T = \sum_{i=1}^n (T_{\ell_i} + T_{m_i})$$

$$T_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \dot{\underline{p}}_i^{*T} \dot{\underline{p}}_i^* \rho dV$$

velocity vector of link
elementary particle:

$$\begin{aligned} \dot{\underline{p}}_i^* &= \dot{\underline{p}}_{\ell_i} + \underline{\omega}_i \times \underline{r}_i \\ &= \dot{\underline{p}}_{\ell_i} + S(\underline{\omega}_i) \underline{r}_i \end{aligned}$$

$$\underline{p}_{\ell_i} = \frac{1}{m_{\ell_i}} \int_{V_{\ell_i}} \underline{p}_i^* \rho dV$$



Computation of the Kinetic Energy

Link i

$$T = \sum_{i=1}^n (T_{\ell_i} + T_{m_i})$$

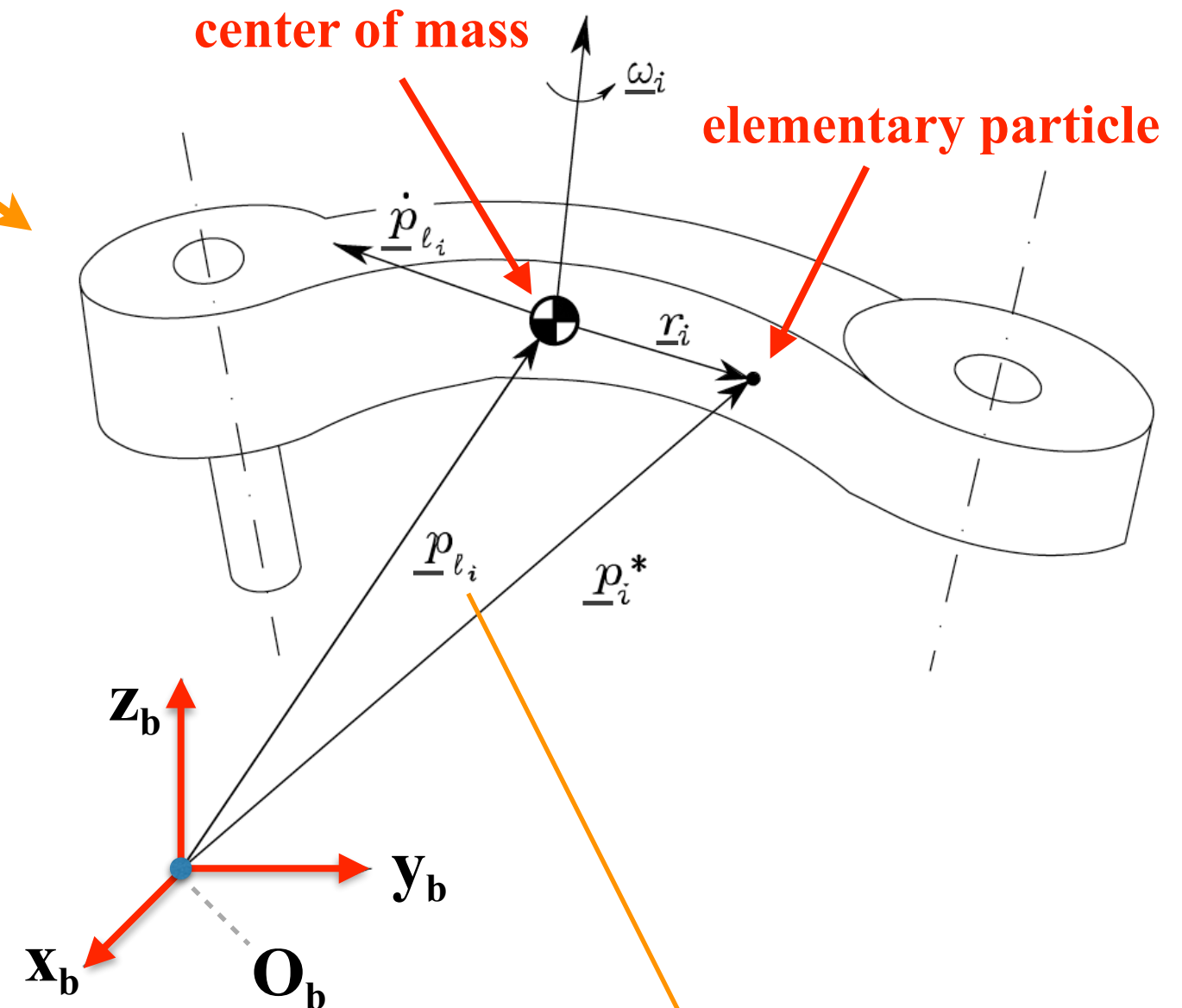
$$T_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{p}}_i^{*T} \underline{\dot{p}}_i^* \rho dV$$

velocity vector of link
elementary particle:

$$\begin{aligned} \underline{\dot{p}}_i^* &= \underline{\dot{p}}_{\ell_i} + \underline{\omega}_i \times \underline{r}_i \\ &= \underline{\dot{p}}_{\ell_i} + \underline{S}(\underline{\omega}_i) \underline{r}_i \end{aligned}$$

linear velocity of the centre of mass of the link

angular velocity of the link



$$\underline{p}_{\ell_i} = \frac{1}{m_{\ell_i}} \int_{V_{\ell_i}} \underline{p}_i^* \rho dV$$

Computation of the Kinetic Energy

The kinetic energy of each link can be subdivided into three contributions:

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

Translational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} \rho dV = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i}$$

Mutual

$$2 \left(\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\boldsymbol{\omega}}_i) \underline{\mathbf{r}}_i \rho dV \right) = 2 \left(\frac{1}{2} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\boldsymbol{\omega}}_i) \int_{V_{\ell_i}} (\underline{\mathbf{p}}_i^* - \underline{\mathbf{p}}_{\ell_i}) \rho dV \right) = 0$$

$$\int_{V_{\ell_i}} \underline{\mathbf{p}}_i^* \rho dV = \underline{\mathbf{p}}_{\ell_i} \int_{V_{\ell_i}} \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV = \frac{1}{2} \underline{\boldsymbol{\omega}}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\boldsymbol{\omega}}_i$$

$$\underline{\dot{\mathbf{p}}}_i^* = \underline{\dot{\mathbf{p}}}_{\ell_i} + \underline{\boldsymbol{\omega}}_i \times \underline{\mathbf{r}}_i = \underline{\dot{\mathbf{p}}}_{\ell_i} + \mathbf{S}(\underline{\boldsymbol{\omega}}_i) \underline{\mathbf{r}}_i$$

$$\mathbf{S}(\underline{\boldsymbol{\omega}}_i) \underline{\mathbf{r}}_i = -\mathbf{S}(\underline{\mathbf{r}}_i) \underline{\boldsymbol{\omega}}_i$$

Computation of the Kinetic Energy

The kinetic energy of each link can be subdivided into three contributions:

Translational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} \rho dV = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i}$$

Mutual

$$2 \left(\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV \right) = 2 \left(\frac{1}{2} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \int_{V_{\ell_i}} (\underline{\mathbf{p}}_i^* - \underline{\mathbf{p}}_{\ell_i}) \rho dV \right) = 0$$

$$\int_{V_{\ell_i}} \underline{\mathbf{p}}_i^* \rho dV = \underline{\mathbf{p}}_{\ell_i} \int_{V_{\ell_i}} \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\omega}_i$$

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \mathbf{I}_{\ell_i} \underline{\omega}_i$$

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV$$

Computation of the Kinetic Energy

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Mutual

$$2 \left(\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV \right) = 2 \left(\frac{1}{2} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \int_{V_{\ell_i}} (\underline{\mathbf{p}}_i^* - \underline{\mathbf{p}}_{\ell_i}) \rho dV \right) = 0$$

$$\int_{V_{\ell_i}} \underline{\mathbf{p}}_i^* \rho dV = \underline{\mathbf{p}}_{\ell_i} \int_{V_{\ell_i}} \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\omega}_i$$

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \mathbf{I}_{\ell_i} \underline{\omega}_i$$

inertia tensor

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV$$

**Inertia tensor: expressed
in the base reference
frame, function of the
joint variables vector**

Computation of the Kinetic Energy

The kinetic energy of each link can be subdivided into three contributions:

Translational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} \rho dV = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i}$$

Mutual

$$2 \left(\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV \right) = 2 \left(\frac{1}{2} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \mathbf{S}(\underline{\omega}_i) \int_{V_{\ell_i}} (\underline{\mathbf{p}}_i^* - \underline{\mathbf{p}}_{\ell_i}) \rho dV \right) = 0$$

$$\int_{V_{\ell_i}} \underline{\mathbf{p}}_i^* \rho dV = \underline{\mathbf{p}}_{\ell_i} \int_{V_{\ell_i}} \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\omega}_i$$

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \mathbf{I}_{\ell_i} \underline{\omega}_i$$

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV$$

Computation of the Kinetic Energy

The kinetic energy of each link can be subdivided into three contributions:

Translational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} \rho dV = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i}$$

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} + \frac{1}{2} \underline{\omega}_i^T \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \underline{\omega}_i$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\omega}_i$$

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \mathbf{I}_{\ell_i} \underline{\omega}_i$$

inertia tensor

Computation of the Kinetic Energy

The kinetic energy of each link can be subdivided into three contributions:

Translational

rotation matrix from **i** to the base reference frame

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} \rho dV = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i}$$

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} + \frac{1}{2} \underline{\omega}_i^T \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \underline{\omega}_i$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} \int_{V_{\ell_i}} \underline{\dot{\mathbf{p}}}_i^{*T} \underline{\dot{\mathbf{p}}}_i^* \rho dV$$

Rotational

$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \left(\int_{V_{\ell_i}} \mathbf{S}^T(\underline{\mathbf{r}}_i) \mathbf{S}(\underline{\mathbf{r}}_i) \rho dV \right) \underline{\omega}_i$$

When referred to the reference frame **i**, the inertia tensor of link **i** is constant

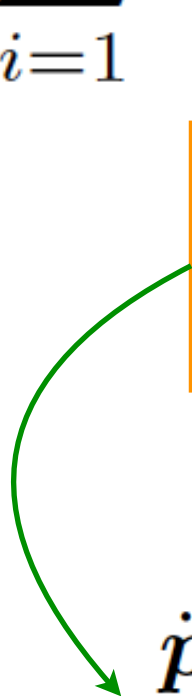
$$\frac{1}{2} \int_{V_{\ell_i}} \underline{\mathbf{r}}_i^T \mathbf{S}^T(\underline{\omega}_i) \mathbf{S}(\underline{\omega}_i) \underline{\mathbf{r}}_i \rho dV = \frac{1}{2} \underline{\omega}_i^T \mathbf{I}_{\ell_i} \underline{\omega}_i$$

inertia tensor

Computation of the Kinetic Energy

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

It is necessary to express the kinetic energy as a function of the joint variables and their first time derivatives. To this end, we can use the geometric approach introduced to describe the differential kinematics of the robot


$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} + \frac{1}{2} \underline{\boldsymbol{\omega}}_i^T \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \underline{\boldsymbol{\omega}}_i$$

$$\begin{aligned} \underline{\dot{\mathbf{p}}}_{\ell_i} &= \mathbf{J}_{P1}^{(\ell_i)} \dot{q}_1 + \dots + \mathbf{J}_{Pi}^{(\ell_i)} \dot{q}_i = \mathbf{J}_P^{(\ell_i)} \underline{\dot{\mathbf{q}}} \\ \underline{\boldsymbol{\omega}}_i &= \mathbf{J}_{O1}^{(\ell_i)} \dot{q}_1 + \dots + \mathbf{J}_{Oi}^{(\ell_i)} \dot{q}_i = \mathbf{J}_O^{(\ell_i)} \underline{\dot{\mathbf{q}}} \end{aligned}$$

$$\begin{aligned} \mathbf{J}_P^{(\ell_i)} &= \begin{bmatrix} \mathbf{J}_{P1}^{(\ell_i)} & \dots & \mathbf{J}_{Pi}^{(\ell_i)} & \mathbf{0} & \dots & \mathbf{0} \end{bmatrix} \\ \mathbf{J}_O^{(\ell_i)} &= \begin{bmatrix} \mathbf{J}_{O1}^{(\ell_i)} & \dots & \mathbf{J}_{Oi}^{(\ell_i)} & \mathbf{0} & \dots & \mathbf{0} \end{bmatrix} \end{aligned}$$

Computation of the Kinetic Energy

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

It is necessary to express the kinetic energy as a function of the the joint variables and their first time derivatives. To this end, we can use the geometric approach introduced to describe the differential kinematics of the robot

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} + \frac{1}{2} \underline{\boldsymbol{\omega}}_i^T \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \underline{\boldsymbol{\omega}}_i$$

Now we need to apply it to an intermediate generic link i instead that to the end-effector

$$\begin{aligned} \underline{\dot{\mathbf{p}}}_{\ell_i} &= \mathbf{J}_{P1}^{(\ell_i)} \dot{q}_1 + \dots + \mathbf{J}_{Pi}^{(\ell_i)} \dot{q}_i = \mathbf{J}_P^{(\ell_i)} \underline{\dot{\mathbf{q}}} \\ \underline{\boldsymbol{\omega}}_i &= \mathbf{J}_{O1}^{(\ell_i)} \dot{q}_1 + \dots + \mathbf{J}_{Oi}^{(\ell_i)} \dot{q}_i = \mathbf{J}_O^{(\ell_i)} \underline{\dot{\mathbf{q}}} \end{aligned}$$

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Computation of the Kinetic Energy

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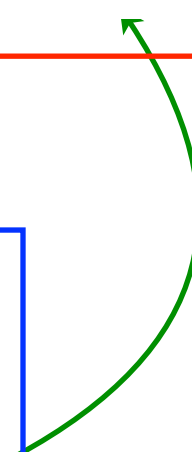
only the contributions of the Jacobian columns up to column i have been considered

Computation of the Kinetic Energy

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$


$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{p}}}_{\ell_i}^T \underline{\dot{\mathbf{p}}}_{\ell_i} + \frac{1}{2} \underline{\boldsymbol{\omega}}_i^T \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \underline{\boldsymbol{\omega}}_i$$


$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \underline{\dot{\mathbf{q}}}^T \mathbf{J}_P^{(\ell_i)T} \mathbf{J}_P^{(\ell_i)} \underline{\dot{\mathbf{q}}} + \frac{1}{2} \underline{\dot{\mathbf{q}}}^T \mathbf{J}_O^{(\ell_i)T} \mathbf{R}_i \mathbf{I}_{\ell_i}^i \mathbf{R}_i^T \mathbf{J}_O^{(\ell_i)} \underline{\dot{\mathbf{q}}}$$


$$\mathbf{J}_P^{(\ell_i)} = \begin{bmatrix} \mathbf{J}_{P1}^{(\ell_i)} & \cdots & \mathbf{J}_{Pi}^{(\ell_i)} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}$$
$$\mathbf{J}_O^{(\ell_i)} = \begin{bmatrix} \mathbf{J}_{O1}^{(\ell_i)} & \cdots & \mathbf{J}_{Oi}^{(\ell_i)} & \mathbf{0} & \cdots & \mathbf{0} \end{bmatrix}$$

Computation of the Kinetic Energy

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$


The kinetic energy contribution of the motor of Joint i can be computed in a way formally analogous to that of the link. I avoid to repeat the procedure, and I copy here directly the initial expression and the final result

$$\mathcal{T}_{m_i} = \frac{1}{2} m_{m_i} \dot{\underline{p}}_{m_i}^T \dot{\underline{p}}_{m_i} + \frac{1}{2} \underline{\omega}_{m_i}^T \mathbf{I}_{m_i} \underline{\omega}_{m_i}$$


$$\mathcal{T}_{m_i} = \frac{1}{2} m_{m_i} \dot{\underline{q}}^T \mathbf{J}_P^{(m_i)T} \mathbf{J}_P^{(m_i)} \dot{\underline{q}} + \frac{1}{2} \dot{\underline{q}}^T \mathbf{J}_O^{(m_i)T} \mathbf{R}_{m_i} \mathbf{I}_{m_i} \mathbf{R}_{m_i}^T \mathbf{J}_O^{(m_i)} \dot{\underline{q}}$$

Computation of the Kinetic Energy

The total kinetic energy of the manipulator with actuators included is given by the quadratic form:

$$\mathcal{T} = \sum_{i=1}^n (\mathcal{T}_{\ell_i} + \mathcal{T}_{m_i})$$

$$\mathcal{T}_{m_i} = \frac{1}{2} m_{m_i} \dot{\underline{q}}^T \mathbf{J}_P^{(m_i)T} \mathbf{J}_P^{(m_i)} \dot{\underline{q}} + \frac{1}{2} \dot{\underline{q}}^T \mathbf{J}_O^{(m_i)T} \mathbf{R}_{m_i} \mathbf{I}_{m_i}^T \mathbf{R}_{m_i}^T \mathbf{J}_O^{(m_i)} \dot{\underline{q}}$$

$$\mathcal{T}_{\ell_i} = \frac{1}{2} m_{\ell_i} \dot{\underline{q}}^T \mathbf{J}_P^{(\ell_i)T} \mathbf{J}_P^{(\ell_i)} \dot{\underline{q}} + \frac{1}{2} \dot{\underline{q}}^T \mathbf{J}_O^{(\ell_i)T} \mathbf{R}_i \mathbf{I}_{\ell_i}^T \mathbf{R}_i^T \mathbf{J}_O^{(\ell_i)} \dot{\underline{q}}$$

Total Kinetic Energy

$$\mathcal{T} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n b_{ij}(\underline{q}) \dot{q}_i \dot{q}_j = \frac{1}{2} \dot{\underline{q}}^T \mathbf{B}(\underline{q}) \dot{\underline{q}}$$

$n \times n$ inertia matrix:

symmetric, positive definite,

and function of the joint variables vector

$$\mathbf{B}(\underline{q}) = \sum_{i=1}^n \left(m_{\ell_i} \mathbf{J}_P^{(\ell_i)T} \mathbf{J}_P^{(\ell_i)} + \mathbf{J}_O^{(\ell_i)T} \mathbf{R}_i \mathbf{I}_{\ell_i}^T \mathbf{R}_i^T \mathbf{J}_O^{(\ell_i)} \right.$$

$$\left. + m_{m_i} \mathbf{J}_P^{(m_i)T} \mathbf{J}_P^{(m_i)} + \mathbf{J}_O^{(m_i)T} \mathbf{R}_{m_i} \mathbf{I}_{m_i}^T \mathbf{R}_{m_i}^T \mathbf{J}_O^{(m_i)} \right)$$



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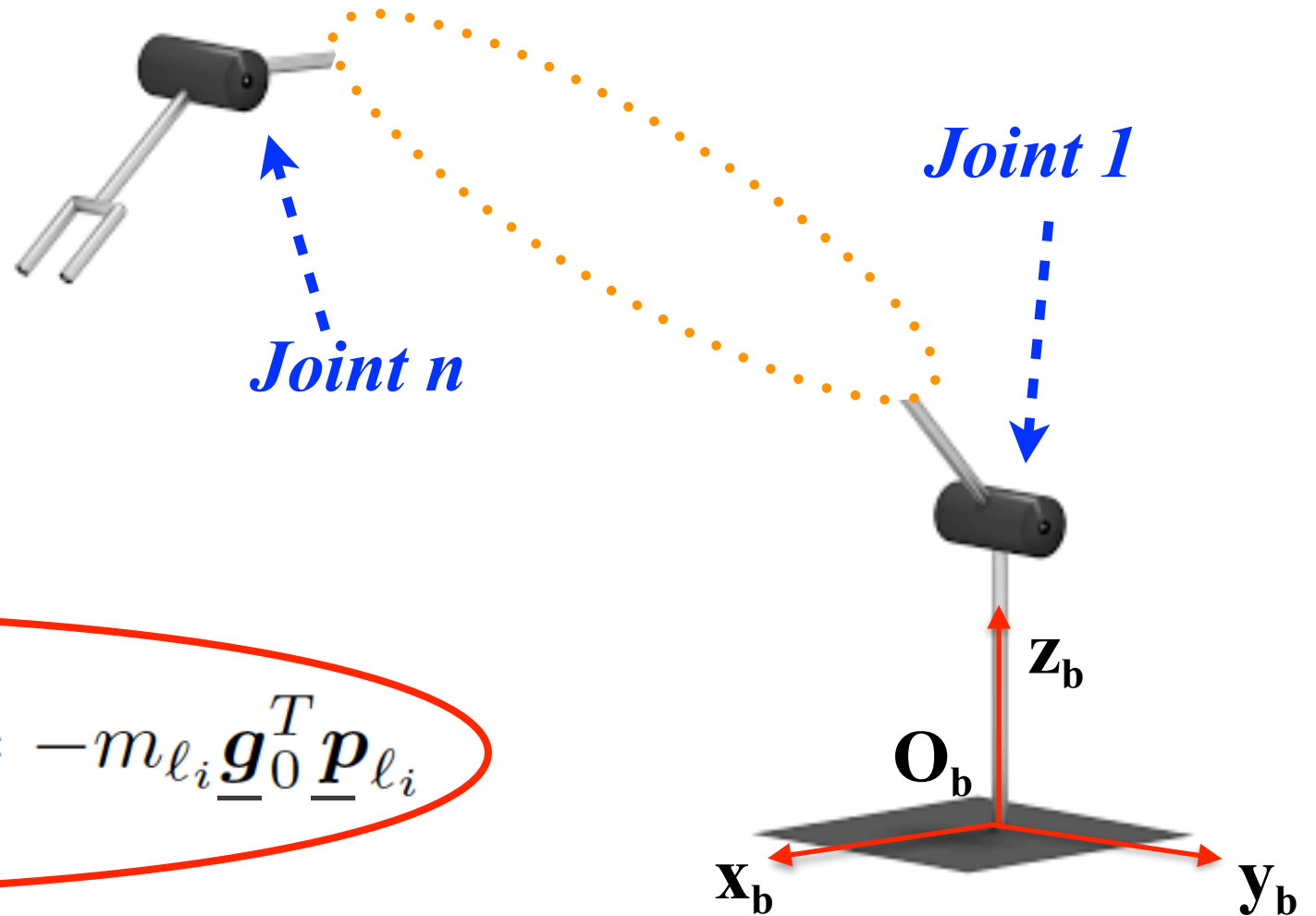


Computation of the Potential Energy

Computation of the Potential Energy

By assuming to have rigid links, the contribution due only to gravitational forces can be expressed as:

$$\mathcal{U} = \sum_{i=1}^n (\mathcal{U}_{\ell_i} + \mathcal{U}_{m_i})$$

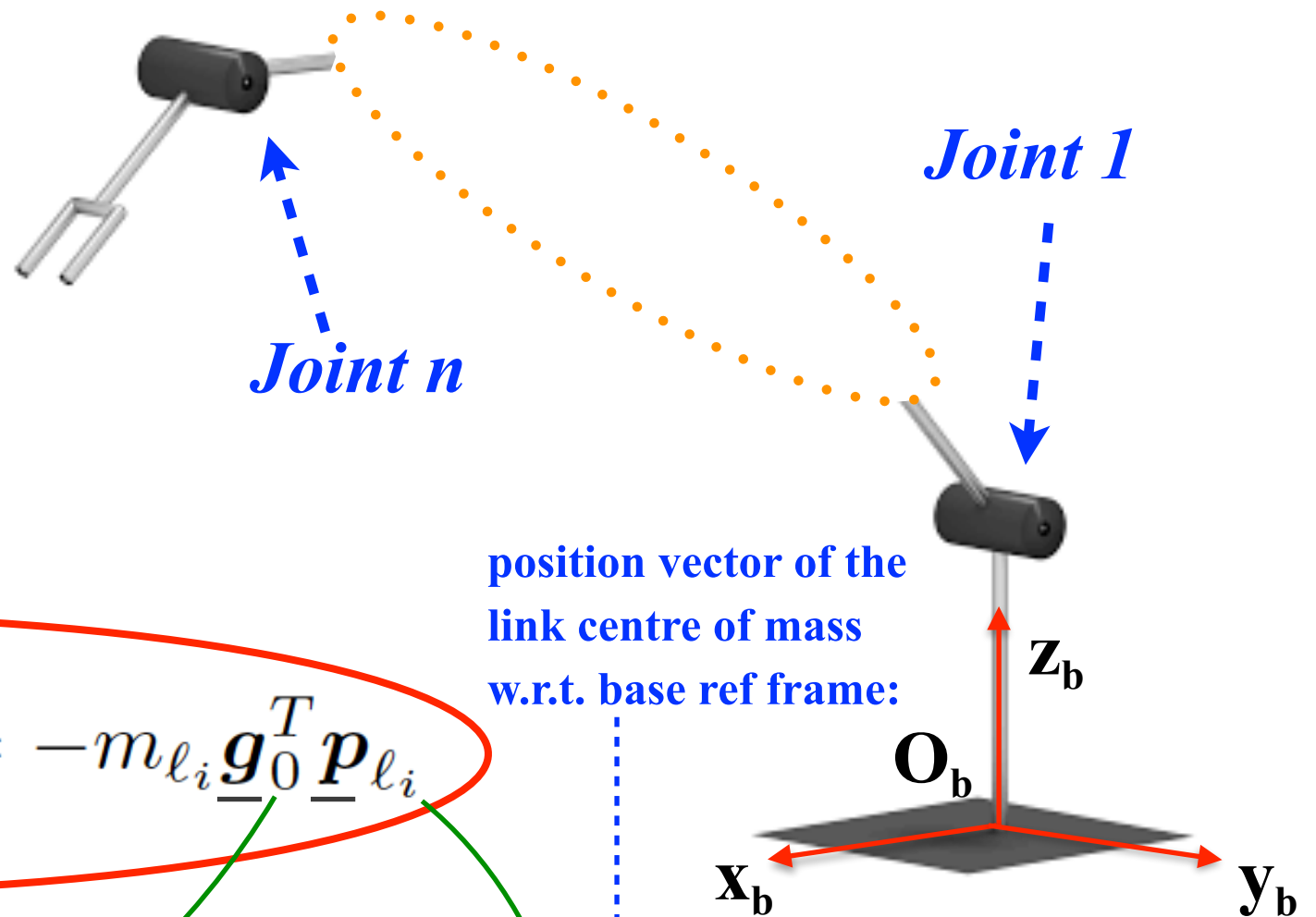


$$\mathcal{U}_{\ell_i} = - \int_{V_{\ell_i}} \underline{g}_0^T \underline{p}_i^* \rho dV = -m_{\ell_i} \underline{g}_0^T \underline{p}_{\ell_i}$$

Computation of the Potential Energy

By assuming to have rigid links, the contribution due only to gravitational forces can be expressed as:

$$\mathcal{U} = \sum_{i=1}^n (\mathcal{U}_{\ell_i} + \mathcal{U}_{m_i})$$



$$\mathcal{U}_{\ell_i} = - \int_{V_{\ell_i}} \underline{g}_0^T \underline{p}_i^* \rho dV = -m_{\ell_i} \underline{g}_0^T \underline{p}_{\ell_i}$$

**gravity acceleration vector
w.r.t. the base reference frame**

$[0 \ 0 \ -g]^T$ since
 z_b is the vertical axis

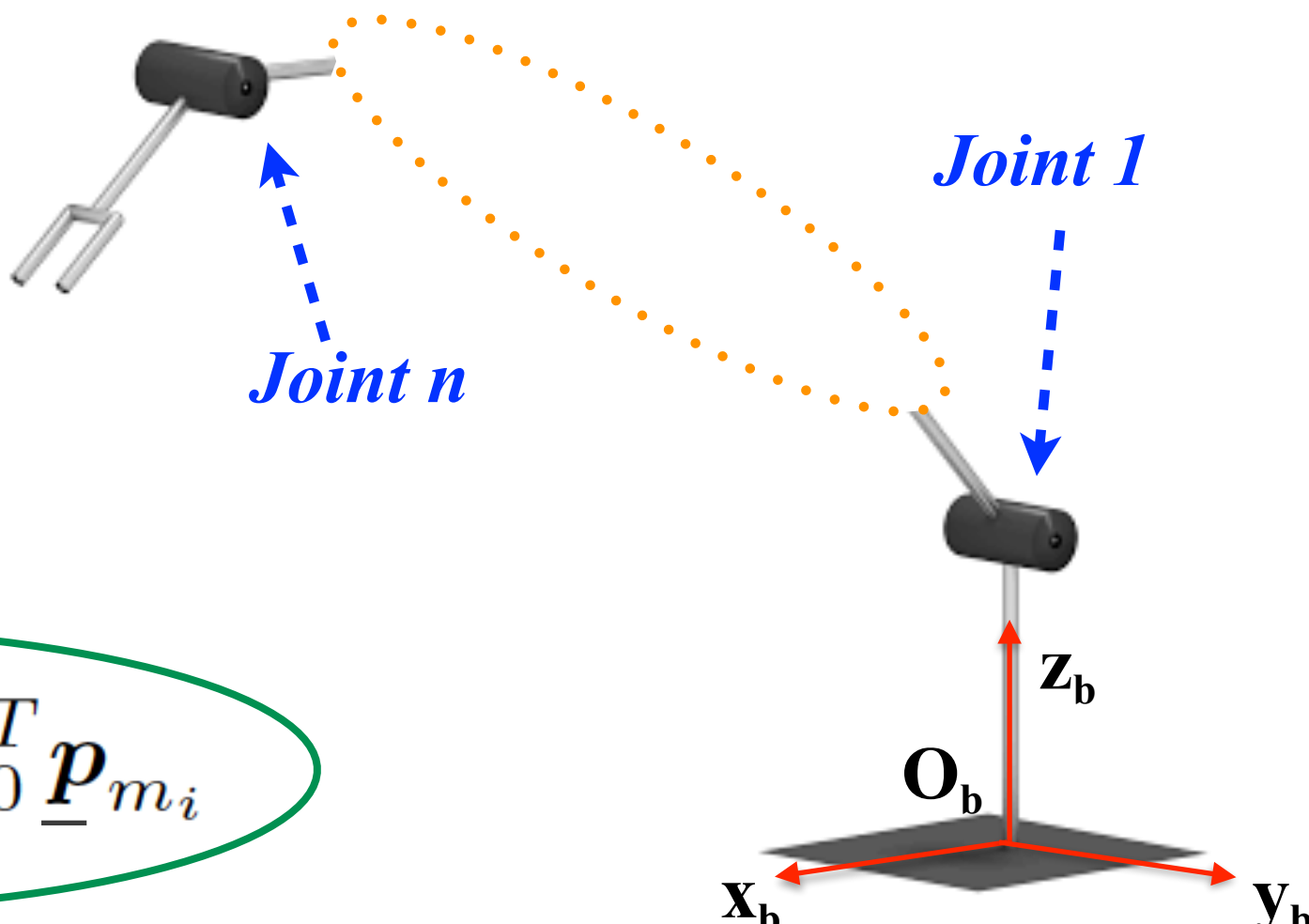
$$\underline{p}_{\ell_i} = \frac{1}{m_{\ell_i}} \int_{V_{\ell_i}} \underline{p}_i^* \rho dV$$

Computation of the Potential Energy

$$\mathcal{U} = \sum_{i=1}^n (\mathcal{U}_{\ell_i} + \mathcal{U}_{m_i})$$

↓

$$\mathcal{U}_{m_i} = -m_{m_i} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_{m_i}$$



The diagram illustrates a robotic arm with two joints, Joint 1 and Joint n, connected by a chain of links. A base coordinate system $(\mathbf{x}_b, \mathbf{y}_b, \mathbf{z}_b)$ is shown at the bottom right, with origin $\mathbf{O}_b$. Dotted orange lines indicate the path of the arm segments. A green arrow points from the summation formula to the potential energy formula for a mass m_i .

Computation of the Potential Energy

$$\mathcal{U} = \sum_{i=1}^n (\mathcal{U}_{\ell_i} + \mathcal{U}_{m_i})$$

$$\mathcal{U}_{m_i} = -m_{m_i} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_{m_i}$$

$$\mathcal{U}_{\ell_i} = - \int_{V_{\ell_i}} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_i^* \rho dV = -m_{\ell_i} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_{\ell_i}$$



Total Potential Energy

$$\mathcal{U} = - \sum_{i=1}^n (m_{\ell_i} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_{\ell_i} + m_{m_i} \underline{\mathbf{g}}_0^T \underline{\mathbf{p}}_{m_i})$$



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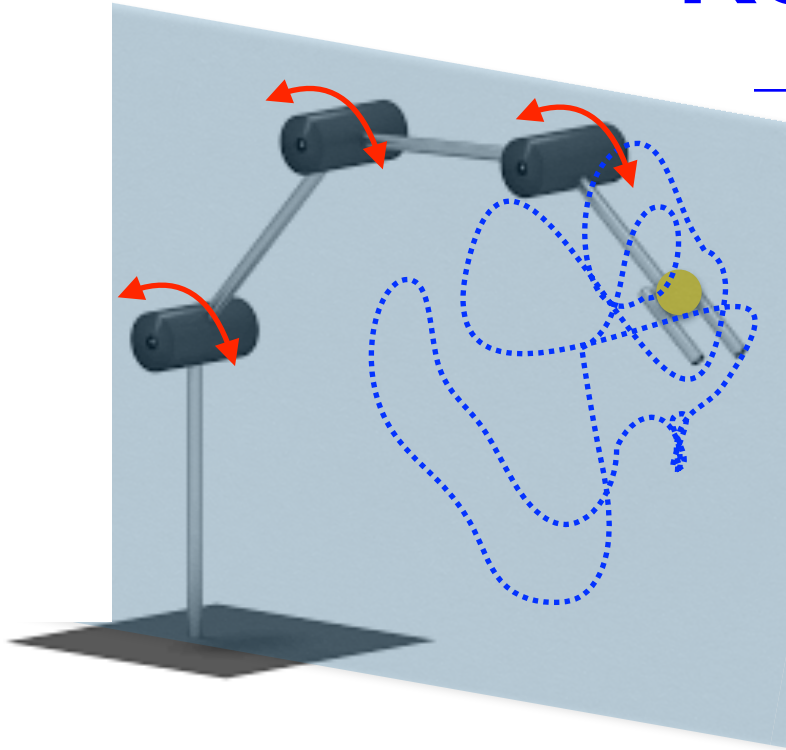
Course of Robot Control

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Robot Dynamic Model

Robot Dynamic Model



Dynamic model of the robot:

collection of the equations of motion of the manipulator in terms of joint variables and their time derivatives up to the second order

**Joints Torques
or Forces**

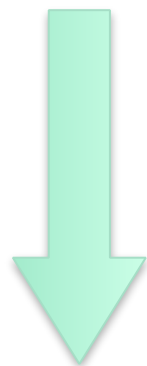


**Motion of the
robot ?**

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$



Robot Dynamic Model

generalized forces applied
to the manipulators

**Joints Torques
or Forces**



**Motion of the
robot ?**

joint positions
joint velocities and
joint accelerations

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

$$\mathcal{L}(\underline{q}, \underline{\dot{q}}) = \mathcal{T}(\underline{q}, \underline{\dot{q}}) - \mathcal{U}(\underline{q})$$

$$\mathcal{T} = \frac{1}{2} \sum_{i=1}^n \sum_{j=1}^n b_{ij}(\underline{q}) \dot{q}_i \dot{q}_j = \frac{1}{2} \underline{\dot{q}}^T \mathbf{B}(\underline{q}) \underline{\dot{q}}$$

$$\mathcal{U} = - \sum_{i=1}^n (m_{\ell_i} \underline{g}_0^T \underline{p}_{\ell_i} + m_{m_i} \underline{g}_0^T \underline{p}_{m_i})$$

generalized forces applied
to the manipulators

**Joints Torques
or Forces**

**Motion of the
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joint positions
joint velocities and
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Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

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Robot Dynamic Model

$$\begin{aligned} \frac{d}{dt} \left(\frac{\partial \mathcal{L}}{\partial \dot{q}_i} \right) &= \frac{d}{dt} \left(\frac{\partial \mathcal{T}}{\partial \dot{q}_i} \right) = \sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \frac{db_{ij}(\underline{q})}{dt} \dot{q}_j \\ &= \sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_{ij}(\underline{q})}{\partial q_k} \dot{q}_k \dot{q}_j \end{aligned}$$

$$\frac{\partial \mathcal{T}}{\partial q_i} = \frac{1}{2} \sum_{j=1}^n \sum_{k=1}^n \frac{\partial b_{jk}(\underline{q})}{\partial q_i} \dot{q}_k \dot{q}_j$$

$$\begin{aligned} \frac{\partial \mathcal{U}}{\partial q_i} &= - \sum_{j=1}^n \left(m_{\ell_j} \underline{\mathbf{g}}_0^T \frac{\partial \underline{\mathbf{p}}_{\ell_j}}{\partial q_i} + m_{m_j} \underline{\mathbf{g}}_0^T \frac{\partial \underline{\mathbf{p}}_{m_j}}{\partial q_i} \right) \\ &= - \sum_{j=1}^n \left(m_{\ell_j} \underline{\mathbf{g}}_0^T \mathbf{J}_{P_i}^{(\ell_j)}(\underline{q}) + m_{m_j} \underline{\mathbf{g}}_0^T \mathbf{J}_{P_i}^{(m_j)}(\underline{q}) \right) = g_i(\underline{q}) \end{aligned}$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model


$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$

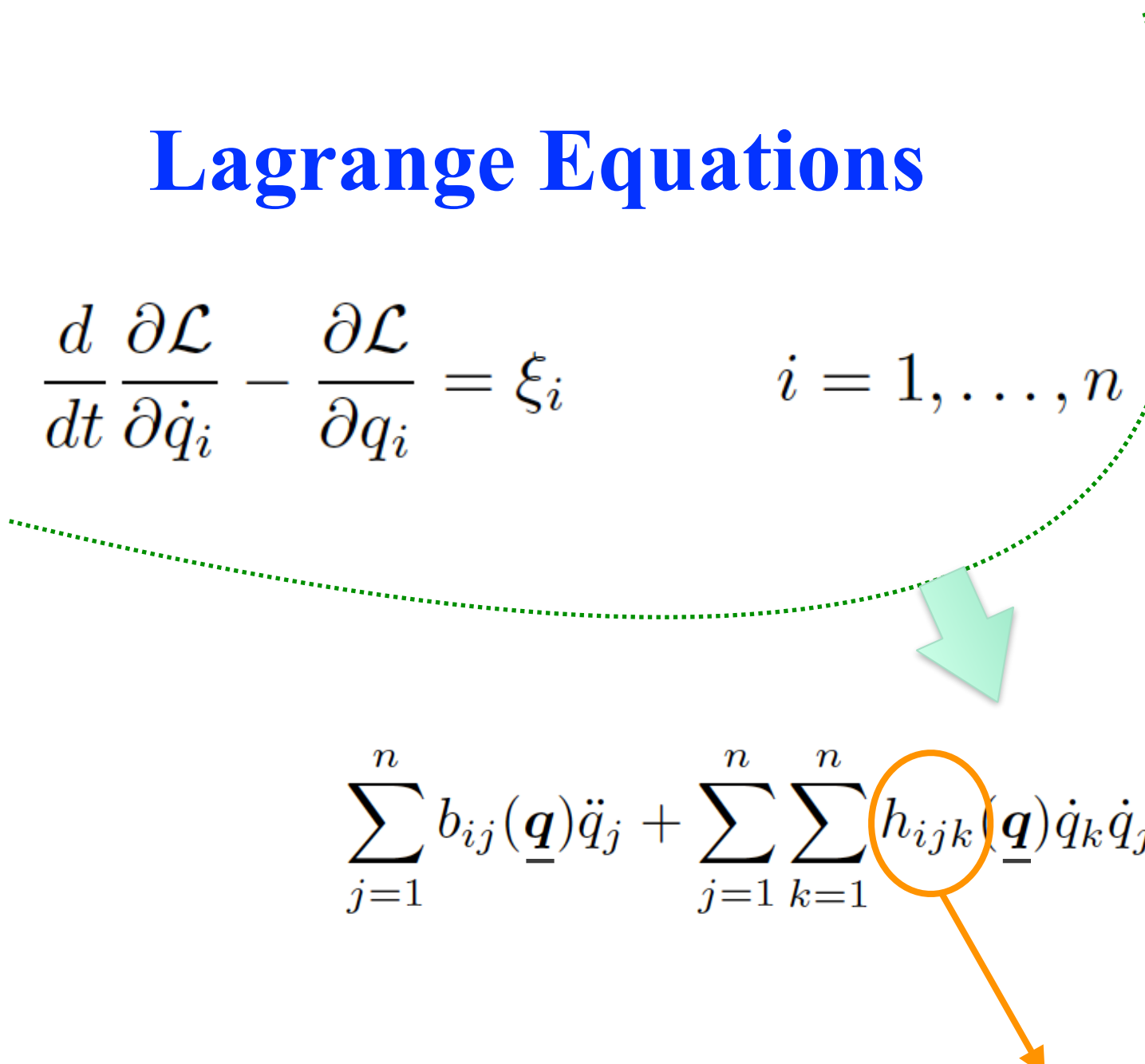
Robot Dynamic Model

Lagrange Equations

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Robot Dynamic Model

$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$


$$h_{ijk} = \frac{\partial b_{ij}}{\partial q_k} - \frac{1}{2} \frac{\partial b_{jk}}{\partial q_i}$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$

they account for the effect of acceleration of joint j on joint i

$$h_{ijk} = \frac{\partial b_{ij}}{\partial q_k} - \frac{1}{2} \frac{\partial b_{jk}}{\partial q_i}$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$

they represent the centrifugal effect induced on joint i by velocity of joint j . Note that $h_{iii}=0$

$$h_{ijk} = \frac{\partial b_{ij}}{\partial q_k} - \frac{1}{2} \frac{\partial b_{jk}}{\partial q_i}$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

it represents the moment generated at joint i axis of the manipulator, in the current configuration, by the presence of gravity

$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$

$$h_{ijk} = \frac{\partial b_{ij}}{\partial q_k} - \frac{1}{2} \frac{\partial b_{jk}}{\partial q_i}$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n b_{ij}(\underline{\mathbf{q}}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{\mathbf{q}}) \dot{q}_k \dot{q}_j + g_i(\underline{\mathbf{q}}) = \xi_i \quad i = 1, \dots, n$$

$$B(\underline{\mathbf{q}}) \ddot{\underline{\mathbf{q}}} + C(\underline{\mathbf{q}}, \dot{\underline{\mathbf{q}}}) \dot{\underline{\mathbf{q}}} + F_v \dot{\underline{\mathbf{q}}} + F_s \operatorname{sgn}(\dot{\underline{\mathbf{q}}}) + \underline{\mathbf{g}}(\underline{\mathbf{q}}) = \underline{\boldsymbol{\tau}} - J^T(\underline{\mathbf{q}}) \underline{\mathbf{h}}_e$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

$n \times n$ inertia matrix:
symmetric, positive definite,
and function of the joint variables vector

$$B(\underline{q}) \ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} + F_v \dot{\underline{q}} + F_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - J^T(\underline{q}) \underline{h}_e$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v \dot{\underline{q}} + F_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - \underline{J}^T(\underline{q})\underline{h}_e$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

the nonconservative forces working
at the manipulator joints

$$B(\underline{q}) \ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}}) \dot{\underline{q}} + \underline{F}_v \dot{\underline{q}} + \underline{F}_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - J^T(\underline{q}) \underline{h}_e$$

Robot Dynamic Model

Lagrange Equations

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}_i} - \frac{\partial \mathcal{L}}{\partial q_i} = \xi_i \quad i = 1, \dots, n$$

Robot Dynamic Model

$$\sum_{j=1}^n c_{ij} \dot{q}_j = \sum_{j=1}^n \sum_{k=1}^n h_{ijk} \dot{q}_k \dot{q}_j$$

$$\sum_{j=1}^n b_{ij}(\underline{q}) \ddot{q}_j + \sum_{j=1}^n \sum_{k=1}^n h_{ijk}(\underline{q}) \dot{q}_k \dot{q}_j + g_i(\underline{q}) = \xi_i \quad i = 1, \dots, n$$

if the end-effector is in contact, a portion of the actuation torques is used to balance the torques induced at the joints by the contact forces

$$B(\underline{q})\ddot{\underline{q}} + C(\underline{q}, \dot{\underline{q}})\dot{\underline{q}} + F_v \dot{\underline{q}} + F_s \operatorname{sgn}(\dot{\underline{q}}) + \underline{g}(\underline{q}) = \underline{\tau} - \underline{J}^T(\underline{q}) \underline{h}_e$$